Number Codes (Set B)

Year 5 Accelerated · **Mark Scheme**

Lesson Worksheet — Answers

Purple — Accessible

1. KPT

2. 754

3. B) OCK

4. GVO

5. 648

6. D) 248

7. AZB

8. 465

9. A) 184, C) 102 and E) 136 are correct (each starts with the digit 1, which codes to T).

10. A) GBE

11. ECU

12. 861

13. C) 421

14. WAR

15. 123

Green — Standard

16. 23 (9 × 2 = 18, 18 + 5 = 23)

17. 8 ((22 + 2) ÷ 3 = 8)

18. C) 13

19. 20 ((6 + 4) × 2 = 20)

20. 10 ((12 − 7) × 2 = 10)

21. A) 7

22. 27 (rule: ×4 then −1; 7 × 4 − 1 = 27)

23. 7 (rule: ×2 then +6; (20 − 6) ÷ 2 = 7)

24. C) 7, D) 8 and E) 9 are correct (codes 22, 25 and 28)

25. E) 38

26. 21 ((10 − 3) × 3 = 21)

27. 6 ((31 − 1) ÷ 5 = 6)

28. B) 28

29. 32 (rule: ×3 then +2; 10 × 3 + 2 = 32)

30. 5 (rule: +1 then ×2; (12 ÷ 2) − 1 = 5)

Red — Challenge

31. 719 (4 → 7, 8 → 11 → 1, 6 → 9)

32. 729 (reverse: 7 → 4, 5 → 2, 2 → 9, giving 729)

33. D) 830

34. 322 (reverse 123 to 321, then + 1 = 322)

35. 554 (reverse of 554 is 455; 455 + 1 = 456)

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36. A) 241 (reverse 241 = 142; 142 − 2 = 140)

37. NIM (471 → M, I, N → reversed → NIM)

38. 629 (reverse PAT to TAP → T, A, P → 6, 2, 9)

39. A) 142, C) 489 and E) 954 are correct — each contains the digit 4, which becomes 0 under this code.

40. B) VKM (5 = K, 7 = M, 3 = V, so 357 → V, K, M)

41. GTD (3 = D, 4 = G, 1 = T, so 413 → G, T, D)

42. 528 (L = 5, W = 2, R = 8 → 528)

43. E) 863

44. 039 (3 → 10 → 0, 6 → 13 → 3, 2 → 9)

45. 479

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Homework Worksheet — Answers

Purple — Accessible

1. BES

2. 235

3. C) SUN

4. RAF

5. 481

6. A) 237

7. TNE

8. 137

9. A) 341, D) 251 and E) 501 are correct (each ends in the digit 1, which codes to T).

10. E) OPT

11. OGL

12. 325

13. B) 256

14. ACT

15. 267

Green — Standard

16. 14 (5 × 2 = 10, 10 + 4 = 14)

17. 8 ((25 − 1) ÷ 3 = 8)

18. D) 19

19. 18 ((4 + 2) × 3 = 18)

20. 16 ((11 − 3) × 2 = 16)

21. C) 7

22. 25 (rule: ×4 then +1; 6 × 4 + 1 = 25)

23. 5 (rule: ×3 then +1; (16 − 1) ÷ 3 = 5)

24. A) 6, B) 7 and C) 8 are correct (codes 15, 17 and 19)

25. A) 33

26. 20 ((5 − 1) × 5 = 20)

27. 7 ((9 + 5) ÷ 2 = 7)

28. B) 17

29. 29 (rule: ×3 then +2; 9 × 3 + 2 = 29)

30. 7 (rule: ×3 then +1; (22 − 1) ÷ 3 = 7)

Standard / Advanced

31. 18 (2 → 6 → 18)

32. 70 (14 → 47 → 70)

33. 28 (original = 20 − 6 = 14; double it = 28)

34. 51 (code 52 means reverse(n) + 1 = 52, so n = 15; reversing 15 alone gives 51)

35. 4 (code 15 means 2n − 1 = 15, so n = 8; half of 8 is 4)

36. A) 315 (391 → reversed 193 → each digit +2 → 315)

37. 391 (315 → subtract 2 from each digit → 193 → reverse → 391)

38. C) 64 (23 × 2 = 46, reversed = 64)

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39. 24 (reverse 84 → 48, then halve → 24)

40. B) 9 (24 × 3 = 72; 7 + 2 = 9)

41. 36 (100 − 37 = 63, reversed = 36)

42. D) 88 (reverse 21 → 12; 100 − 12 = 88)

43. 61 (rule: ×2 then +1; 30 × 2 + 1 = 61)

44. B) 23 and D) 8 are correct (23 → 46 → 64; 8 → 16 → 61; both exceed 50)

45. 18 ((n − 7) × 2 = 22, so n − 7 = 11, n = 18)

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